

Quantization-Aware Digital PID Control of an Inverting Buck-Boost Converter: A Simulation-to-Hardware Feasibility Audit on ESP32-S3 and STM32F103

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ABSTRACT Digital control of switching power converters is shaped by three implementation-quantized quantities: the ADC resolution of the sensed output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized. Whether the resulting design limits are properties of the hardware or of the software stack is rarely asked. This feasibility audit asks that question for an inverting buck-boost converter ($V_{ref} = -12\text{ V}$, $f_{sw} = 100\text{ kHz}$). The paper plant is ideal ($L = 100\mu\text{H}$, $C = 220\mu\text{F}$, $R = 10\Omega$, $R_L = \text{ESR} = 0$, $f_0 \approx 1.07\text{ kHz}$); the as-wired hardware plant is $5\Omega + 330\mu\text{H} + 2.2\mu\text{F}$ ($f_0 \approx 5.9\text{ kHz}$, $Q \approx 2.45$) plus LM358 follower. A 79-run switched-model study on the paper plant with zero delay first locates the critical sampling boundary near $3.5\text{--}4 \times T_{sw}$ ($35\text{--}40\mu\text{s}$) and shows the three quantizers interact super-additively; delay- ($65\mu\text{s}$ ESP Arduino, $11\mu\text{s}$ STM32 bare) and noise- ($\sigma \approx 1.65/\sqrt{N}\text{ V}$) augmented re-runs explain why the $2.2\mu\text{F}$ hardware never settles with frozen gains even at $1 \times$ (Fig. 20). The same controller is ported, from one shared firmware source, to ESP32-S3 (240 MHz, FPU) and STM32F103 (72 MHz, no FPU). The central finding is an abstraction-layer tax: through Arduino both MCUs are ADC-bound (ESP32-S3 $60\mu\text{s}$, $73\mu\text{s}$ floor $7.3 \times T_{sw}$; STM32 $72.8\mu\text{s}/23.9\mu\text{s}$, $232\mu\text{s}$ soft-float quantizer), but STM32 bare recovers $11.0\mu\text{s}$ ADC (first-sample $97\mu\text{s}$, $17 \times$ ideal) and $39.8\mu\text{s}$ paced floor ($4.0 \times$) vs ESP $70.8\mu\text{s}$ — still inside the paper-plant stable region while ESP is deep outside. Baseline hardware is pacing/noise metrology on the jumper surrogate; closed-loop as-wired validation (§V-G) shows Mode T (paced, single raw, frozen gains) never settles on the LC (447ms window-clip) while Mode S (averaged $N = 1 + \text{EMA}$) settles at 3.6ms, and an adaptive $N = 8 \leftrightarrow 64$ demo on the surrogate cuts updates 84% (0.98 on the LC) while holding $\pm 40\text{ mV}$. All data, firmware and $220\mu\text{F}$ vs $330\mu\text{H}$ simulations are in the companion repo.

INDEX TERMS digital PID control, quantization, buck-boost converter, ESP32-S3, STM32F103, cross-platform, adaptive sampling, measurement latency

I. INTRODUCTION

Digital control loops are now standard in switching power converters, driven by programmability, immunity to analog component drift, and integration with digital loads [1], [2]. The complete digital loop contains two unavoidable quantizers: the analog-to-digital converter (ADC) that samples the output voltage, and the digital pulse-width modulator (DPWM) that resolves the duty ratio. Their resolutions, together with the sampling period of the controller, jointly determine regulation

quality. Classic work established that coarse quantization causes steady-state limit cycling and conditions between DPWM and ADC resolution for its elimination [3], [4], [5], while quantization-effect modeling permits quantitative prediction of these oscillations [6]. Recent work extends the analysis to MHz-range GaN converters using dither and sigma-delta modulation [7], [8], [9] and to design guidelines for two-loop current-mode digital controllers [8].

Digital control relieves the analog constraints of pas-

sive compensation, enables programmability and auto-tuning, and integrates naturally with digital loads [1], [2]. In exchange, the designer accepts three implementation-quantized quantities that an analog loop never sees: the ADC resolution of the measured output, the DPWM resolution of the commanded duty, and the sampling period at which controller and power stage are synchronized; DPWM-architecture and delay considerations are central to the achievable resolution [10], [11].

The theoretical basis is well established. Peterchev and Sanders derived conditions on ADC/DPWM resolution trade-offs for limit-cycle elimination and introduced digital dither to recover effective DPWM resolution [3]. Peng, Prodić, Alarcón, and Maksimović derived static and dynamic quantization models explaining the origins of limit-cycle oscillations [4], and Maksimović and Zane provided the small-signal discrete-time framework used to design digitally controlled converters [6]. A fuller survey appears in [12]. On the application side, low-resolution non-uniform ADC quantization has been shown to improve dynamic response [13], and the full control loop benefits from predictive structures [14], [15].

Two observations motivate this work. First, most quantization studies isolate a single stage, whereas a practical controller is constrained in all three quantities simultaneously; the interaction between sampling period, ADC word length, and PWM word length is rarely treated as one design problem. Second, the buck-boost converter, whose inverted output and non-minimum phase control path [16] make regulation inherently harder, is underrepresented in the digital-control literature: most buck-boost-specific digital studies consider time-domain or PID structures at a single resolution [17], [18] and do not sweep the full implementation space. The classical converter model itself is textbook material [19].

The central question of this work is therefore: how do sampling period, ADC quantization, PWM resolution, and real-time computational latency jointly affect the closed-loop performance of a microcontroller-controlled inverting buck-boost converter, and can a lightweight quantization-aware adaptive PID maintain acceptable regulation while reducing unnecessary controller-update effort — and do the answers depend on the microcontroller? Answering it requires three levels of validation that earlier studies treat separately: a continuous (analog) baseline, a digital simulation that introduces sampling, ADC and PWM quantization and computational delay, and surrogate (jumper-emulated) measurements on two embedded platforms. The microcontrollers are not merely devices that run the PID; they are the experimental vehicles for exposing quantities a simulator cannot produce directly: the acquisition time of the on-chip ADC, the execution time of the control step (with and without a hardware FPU), the update latency of the PWM peripheral, and the resulting control-loop latency and CPU utilization. The study uses one shared

firmware source on an ESP32-S3 (240 MHz, Arduino core, LEDC + analogRead) and an STM32F103 Blue Pill (72 MHz, no FPU, STM32duino), whose $17\times$ bare-register ADC advantage isolates whether the predicted sampling boundary is a hardware fact or a firmware artifact.

This paper reports a simulation-to-hardware feasibility audit of a digitally controlled inverting buck-boost converter under simultaneous quantization of sampling period, ADC resolution, and PWM resolution. The scope is explicitly that of an audit, not a completed converter regulation claim: the simulated plant is ideal and zero-delay, and hardware results are pacing and acquisition-noise metrology on a jumper-emulated first-order surrogate; LC closed-loop regulation is deferred. Contributions: (1) a 79-run factorial study that separates the individual effect of each quantizer from their combined effect, locates the critical sampling boundary near $3.5\text{--}4\times T_{\text{sw}}$ ($35\text{--}40\mu\text{s}$), and quantifies the super-additive interaction (window-clip limited when never settled); (2) an objective worst-case selection made after the sweep, not a priori; (3) the identification of a half-scale ADC coincidence ($V_{\text{ref}} = -\text{FS}/2$) that masks steady-state error at low ADC resolutions; (4) an ESP32-S3 latency decomposition (ADC $60.0\mu\text{s}$ avg, PID $0.86\mu\text{s}$ int32, PWM $4.2\mu\text{s}$, $T_{\text{lat}} \approx 65\mu\text{s}$, paced floor $73\mu\text{s} \approx 7.3\times T_{\text{sw}}$); (5) a cross-platform replication on an STM32F103 through both the Arduino layer (ADC $72.8\mu\text{s}$, PWM $23.9\mu\text{s}$, float+quantizer $232\mu\text{s}$ soft-float) and bare-register access (ADC $4.23\mu\text{s}$, PWM $5.88\mu\text{s}$), revealing a $17\times$ ADC tax and the soft-float cost that vanishes with fixed-point/FPU (STM32 logs available on request); (6) a bare-register pacing sweep (mode T, $1\text{--}5\times$, $10\text{--}50\mu\text{s}$) that validates the achievable schedule crosses the simulated boundary on STM32 bare ($26\mu\text{s} \approx 2.6\times T_{\text{sw}}$) while ESP32-S3 cannot, without yet claiming closed-loop regulation across it; (7) an effective-resolution experiment showing the $1.6\text{--}1.9\text{V}$ acquisition-noise floor masks the simulated quantization effects at achievable schedules; (8) a surrogate adaptive schedule+dither demo ($N = 8$ vs 64 , update-reduction 84% , $\pm 40\text{mV}$, $n = 1$) validating the resource-aware trade-off on the feasible regime; and (9) delay- and noise-augmented re-simulations that inject the measured T_{lat} and $\sigma_{\text{ADC}} \approx 1.65/\sqrt{N}\text{V}$ to reconcile simulation with hardware. Novelty is the integration and experimental stack-tax characterization from one shared firmware source, not variable-rate sampling or dithering alone. Hardware closed-loop validation with a real LC plant is software-ready and deferred to the next revision.

II. SYSTEM MODEL AND DIGITAL CONTROL

The paper plant is an ideal inverting buck-boost converter (Fig. 1) with input $V_{\text{in}} = +12\text{V}$, reference $V_{\text{ref}} = -12\text{V}$, $L = 100\mu\text{H}$, $C = 220\mu\text{F}$, switching frequency $f_{\text{sw}} = 100\text{kHz}$ ($T_{\text{sw}} = 10\mu\text{s}$), and load

$R = 10\ \Omega$ ($f_0 \approx 1.07\ \text{kHz}$). The as-wired hardware plant validated in §V-G is $5\ \Omega + 330\ \mu\text{H} + 2.2\ \mu\text{F}$ ($f_0 \approx 5.9\ \text{kHz}$, $Q \approx 2.45$) plus LM358 follower (Fig. 8; design showed $100\ \mu\text{H}$, $f_0 \approx 10.7\ \text{kHz}$, $Q \approx 1.3$). The $2.2\ \mu\text{F}$ plant is $5\text{--}10\times$ faster than the paper plant and, as §VI and Fig. 20 show, its $35\text{--}40\ \mu\text{s}$ boundary does not transfer without retuning.

During OFF, the inductor current is held at zero when it would otherwise reverse (zero-current boundary), which counts discontinuous-mode (DCM) cycles and keeps the model valid in DCM without clamping the output. Many simplified studies go silent on this point. All reported results are obtained on this switched-state model, not on an averaged approximation, so quantization artifacts (limit cycles, DCM chatter) appear faithfully; nonideal DCM of the buck-boost requires explicit boundary modeling [20].

$$\text{ON: } \frac{di_L}{dt} = \frac{+V_{in}}{L}, \quad \frac{dv_C}{dt} = -\frac{v_C}{RC} \quad (1)$$

$$\text{OFF: } \frac{di_L}{dt} = \frac{v_C}{L}, \quad \frac{dv_C}{dt} = -\frac{i_L + v_C/R}{C} \quad (2)$$

A. DIGITAL CONTROLLER AND QUANTIZATION

A velocity-form digital PID with positive gains (Fig. 2) operates at the sampling period T_s :

$$d[k] = d[k-1] + K_p(e[k] - e[k-1]) + K_i T_s e[k] + \frac{K_d}{T_s}(e[k] - 2e[k-1] + e[k-2]) \quad (3)$$

with $e[k] = v_{\text{meas}}[k] - V_{\text{ref}}$, duty saturated to $[0.05, 0.95]$, and no separate integrator state (inherent anti-windup). The measurement, duty, and update cadence are each quantized independently:

- Sampling: the controller updates every $T_s \in \{T_{\text{sw}}, 2T_{\text{sw}}, 5T_{\text{sw}}\}$ ($10, 20, 50\ \mu\text{s}$);
- ADC: mid-tread quantizer, full-scale $FS = 24\ \text{V}$, $12/10/8/6$ bits. Note $V_{\text{ref}} = -FS/2$, so the reference is exactly representable at every resolution;
- PWM: duty quantized to 2^b levels, $b \in \{12, 10, 8, 6\}$.

A continuous-ideal baseline recomputes the same PID every dt with no quantization (true analog reference). The frozen controller gains were obtained once, deterministically, before any quantization study: grid search over $K_p \in \{0.005, 0.01, 0.02, 0.05\}$, $K_i \in \{20, 100, 500, 2000\}$, $K_d \in \{0, 2 \times 10^{-6}, 10^{-5}\}$ with cost $J = o_v/10 + t_s/5\ \text{ms} + \text{IAE}/0.05$ under constraints $\text{overshoot} \leq 10\%$ and $|e_{\text{ss}}| \leq 60\ \text{mV}$, re-validated on the switched model: $K_p = 0.05$, $K_i = 100.0$, $K_d = 10^{-5}$. No parameter was touched during the quantization study.

III. SIMULATION FRAMEWORK AND METHODOLOGY

The core 79 simulations are on the paper plant ($100\ \mu\text{H}$, $220\ \mu\text{F}$, $10\ \Omega$, $f_0 \approx 1.07\ \text{kHz}$). They comprise: the continuous baseline, the ideal digital controller ($T_s = T_{\text{sw}}$, 16-bit ADC, 16-bit PWM), three sampling-only points, a six-point focused sampling sweep across the critical boundary, four ADC-only, four PWM-only runs, the full $3 \times 4 \times 4$ factorial sweep (48 runs), six disturbance runs (120 ms total, step at $t = 60\ \text{ms}$, evaluated over the 60 ms post-step window, $6000\ T_{\text{sw}}$, on the continuous baseline, the ideal digital controller, and the objectively selected worst configuration), and six operating-point robustness runs ($V_{\text{in}} = 10, 15\ \text{V}$ on three configurations). Audit extensions re-run the same controller on the as-wired plant ($330\ \mu\text{H}$, $2.2\ \mu\text{F}$, $5\ \Omega$, $f_0 \approx 5.9\ \text{kHz}$) and with injected pipeline delay ($65\ \mu\text{s}$ ESP Arduino, $11\ \mu\text{s}$ STM32 bare) and acquisition noise ($\sigma \approx 1.65/\sqrt{N}\ \text{V}$) via `sim_core.BuckBoost(comp_delay_s, adc_noise_sigma)`; see §VI, §VI-A and §V-G. Solver: fixed-step Euler $dt = 1/(f_{\text{sw}} \cdot 200) = 0.05\ \mu\text{s}$ ($200\ \text{steps}/T_{\text{sw}}$), ZOH at period start, zero delay in the 79-run core, $r_{\text{DCM}} = N_{\text{DCM}}/N_{\text{sw}} \leq 1$ asserted.

Metrics: 2% settling time, overshoot, steady-state error, peak-to-peak ripple (limit cycle), IAE, ITAE, and N_{DCM} (switching periods containing a DCM event, $r_{\text{DCM}} = N_{\text{DCM}}/N_{\text{sw}} \leq 1$). For disturbance runs all metrics are evaluated on the 60 ms post-step window ($6000\ T_{\text{sw}}$ within a 120 ms total run) and ripple on its final 50%. The worst configuration is selected objectively after the sweep, as the run with the largest normalized degradation of steady-state error, IAE, and limit cycle relative to the ideal digital controller. No configuration is pre-declared worst.

IV. RESULTS

A. CONTINUOUS VERSUS IDEAL DIGITAL

The conversion from analog to ideal digital control already costs measurable quality (Table 2): the 16-bit/one-cycle system settles 36% slower (2.66 vs 1.96 ms) and shows a 12.76 mV steady-state offset and 39.6 mV peak-to-peak ripple against the continuous 0.44 mV / 26.7 mV. The 12.8 mV residual offset is sampling/ZOH and numerical error for these gains (sampling dominates at 16-bit, not quantization alone); every further result is expressed relative to it.

B. INDIVIDUAL STAGE EFFECTS

Fig. 3 isolates the sampling stage at 16-bit resolution and locates the critical boundary. Settling degrades smoothly as T_s/T_{sw} rises from 1 to 3.5 (2.66 to 6.38 ms, all still regulation-valid), then the loop loses regulation outright at $4\times$: it never settles, the peak-to-peak ripple jumps to 1.59 V, and 623 switching periods containing a DCM event accrue in the 60 ms window ($6000\ T_{\text{sw}}$, $r_{\text{DCM}} = 0.10$). The practical boundary therefore sits

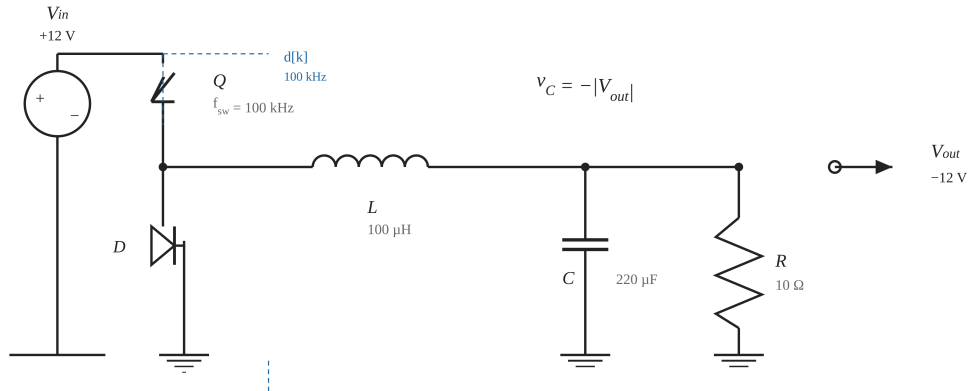


FIGURE 1. Inverting buck-boost power stage used in the study ($V_{in} = +12\text{ V}$, $L = 100\text{ }\mu\text{H}$, $C = 220\text{ }\mu\text{F}$, $f_{sw} = 100\text{ kHz}$, $R = 10\text{ }\Omega$; output $V_{out} \approx -12\text{ V}$ at $D = 0.5$).

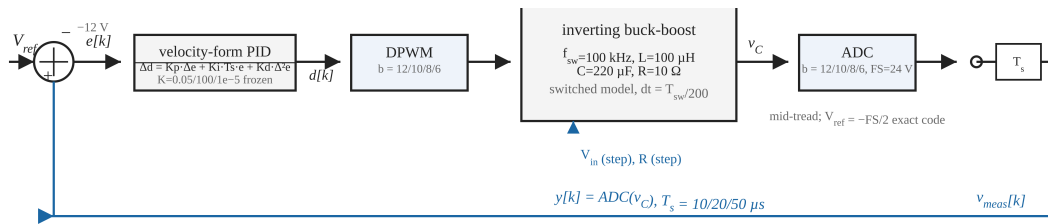


FIGURE 2. Digital control loop: velocity-form PID, DPWM and ADC quantizers ($b \in \{12, 10, 8, 6\}$ bit, $FS = 24\text{ V}$), and sampling at $T_s = 10/20/50\text{ }\mu\text{s}$. The reference sits exactly at the half-scale code $V_{ref} = -FS/2$.

Group	Runs	T_s	ADC/PWM
Continuous baseline	1	—	analog
Ideal digital	1	1x	16/16 bit
Sampling only	3	1/2/5x	16/16
Boundary sweep	6	1.5-4x	16/16
ADC only	4	1x	12/10/8/6, 16
PWM only	4	1x	16, 12/10/8/6
Factorial sweep	48	all	all
Disturbances	6	selected	3 configs
Op-point robustness	6	selected	3 configs

TABLE 1. Study layout (79 simulations).

	Continuous	Ideal digital
Settling (ms)	1.96	2.66
Overshoot (%)	0.07	0.02
e_ss (mV)	0.44	12.76
Limit cycle (mV)	26.7	39.6
IAE	0.0058	0.0109
DCM cycles	0	0

TABLE 2. Continuous baseline vs ideal digital.

between $3.5\times$ and $4\times T_{sw}$ ($35\text{--}40\ \mu\text{s}$); $T_s = 5T_{sw}$ ($50\ \mu\text{s}$) fails in every configuration, alone or combined.

ADC resolution couples to the reference in a subtle way (Fig. 4): because $V_{ref} = -FS/2$ is exactly representable, the 8-bit and 6-bit ADCs lock onto the exact code and reduce the measured steady-state error to $\approx 0.44\text{ mV}$ while masking a worse transient (overshoot rises from 0.06% at 12-bit to 0.85% at 6-bit; settling from 2.70 to 3.48 ms). This is a half-scale coincidence, not an ADC benefit.

PWM resolution below 8 bits dominates the limit cycle (Fig. 5): 6-bit PWM adds a 204.8 mV ripple versus $\approx 39.5\text{ mV}$ at 12/10/8-bit. Duty quantization is the direct source of steady-state ripple, as predicted by limit-cycle analysis [3], [4]; at 6-bit the duty grid step is 1 LSB = $1/64$ over $[0, 1]$ (clamped to $[0.05, 0.95]$; over the clamped 0.9 range this would be $0.9/63 \approx 0.0143$ and 0.69 V static, 9% lower), and the loop settles into a limit cycle toggling between the adjacent codes 31/64 and 32/64 around the $D \approx 0.5$ operating point, whose measured peak-to-peak output excursion is 204.8 mV. The static code-to-code shift at this operating point is larger, $|dV_{out}/dD| \cdot 1/64 = 0.75\text{ V}$ with $|dV_{out}/dD| = V_{in}/(1-D)^2 = 48\text{ V}$ per unit duty at $D = 0.5$; the dynamic ripple is smaller because the output capacitor averages the excursion over the limit-cycle period. This is precisely the regime addressed by finer DPWM controller implementations [11], [10] and by noise-shaping modulators [21].

C. COMBINED SWEEP AND INTERACTIONS

In the 48-run factorial sweep the worst configuration is $T_s = 5T_{sw}$, ADC=8-bit, PWM=12-bit (normalized degradation score 1309.2): settling never completes within 60 ms, steady-state error -13.43 V , peak-to-peak ripple 7.33 V, IAE 0.789 V·s, 884 switching periods containing a DCM event ($r_{DCM} = 0.15$, $6000 T_{sw}$). Fig. 6 classifies every combination against explicit thresholds (settling $\leq 6\text{ ms}$, $|e_{ss}| \leq 120\text{ mV}$, limit cycle $\leq 120\text{ mV}$, overshoot, DCM) into excellent / acceptable / marginal / unusable. Sampled at $T_s = T_{sw}$, 13 of 16 combinations are excellent and the rest acceptable; at $2\times$ all are acceptable or marginal; at $5\times$ all 16 are unusable.

The worst configuration is far worse than any individual degradation: -13.43 V versus the -13.07 V of sampling alone (sampling $5\times$ alone: $N_{DCM} = 789$, $r = 0.13$), but with $N_{DCM} = 884$ ($r = 0.15$) and a 7.33 V peak-to-peak ripple that the sampling-only run does not exhibit at that magnitude. The super-additivity is quantitative with $D = \frac{1}{3} \left[\frac{|m_1 - m_{1,0}|}{\max(|m_{1,0}|, \epsilon_1)} + \frac{|m_2 - m_{2,0}|}{\max(|m_{2,0}|, \epsilon_2)} + \frac{|m_3 - m_{3,0}|}{\max(|m_{3,0}|, \epsilon_3)} \right]$ for metrics $(m_1, m_2, m_3) = (|e_{ss}|, \text{IAE}, \text{LC})$ relative to the ideal digital controller, $m_{j,0} = (12.76\text{ mV}, 0.0109, 39.6\text{ mV})$, $\epsilon = (1\text{ mV}, 10^{-6}, 10^{-3}\text{ mV})$: the steady-state term enters as an absolute difference, IAE and limit cycle as signed changes (a stage that improves them scores lower). The individual terms are $D_{\text{sampling}}(5\times) = 380.5$,

$D_{\text{ADC}}(8\text{-bit}) = 0.20$, $D_{\text{PWM}}(12\text{-bit}) = 0.0$ (unnormalized 1141.6/0.6/0.0, $3\times$ larger); the additive prediction is 380.7 (unnormalized 1142.2), whereas the measured combined degradation is $D_{\text{total}} = 436.4$ (unnormalized 1309.2), an interaction of $+55.7$ (unnormalized $+167$, $+15\%$). When the loop never settles t_{settle} is clipped to the 60 ms window (59.99975 ms) so D is a window-clip ceiling, not a physical settling time; the $+15\%$ therefore marks a saturation regime. The 8-bit ADC coincidence, harmless alone, converts a slowly sampling loop into deep limit-cycle chatter once the 12-bit PWM pulls the operating point across the coincidence.

D. DISTURBANCE REJECTION

Table 3 summarizes the six disturbance runs (120 ms total, step at $t = 60\text{ ms}$; 60 ms post-step window, 6000 T_{sw} ; recovery time = $t_{\text{settle}} - 60\text{ ms}$; ripple on final 50% of post-step). Both healthy controllers reject input and load steps in under 1.5 ms with negligible IAE (IAE $\approx 0.001\text{--}0.002\text{ V}\cdot\text{s}$). The worst configuration never re-regulates within the observation window: it settles into deep DCM chatter with $>5\text{ V}$ of steady-state error, accruing 1750 switching periods with a DCM event after the V_{in} step and 1092 after the load step within the 60 ms post-step window (≤ 6000 , vs 884 undisturbed) — because the loop never settles and keeps crossing the zero-current boundary for the entire window (Fig. 7).

E. OPERATING-POINT ROBUSTNESS

The main conclusions are not an artifact of the single nominal point ($V_{in} = 12\text{ V}$, $V_{ref} = -12\text{ V}$, $R = 10\ \Omega$). Table 4 repeats the ideal, a representative good ($1\times$, 10-bit, 10-bit), and a representative poor ($2\times$, 6-bit, 6-bit) configuration at $V_{in} = 10$ and 15 V with the same frozen gains. Ideal and good configurations retain full regulation (2.5–2.7 ms settling, sub-100 mV limit cycle) across the input range. The poor configuration holds at 10 and 12 V but degrades sharply at $V_{in} = 15\text{ V}$. It settles only at the very edge of the 60 ms window (59.2 ms), with a 561 mV limit cycle and 7 switching periods with a DCM event. Marginal designs are therefore operating-point sensitive, not merely sampling-sensitive.

V. EXPERIMENTAL CLOSED-LOOP PLATFORMS

Two platforms share one firmware source (esp32_feas/src/main.cpp, #ifdef ARDUINO_ARCH_ESP32 hardware layer, modes B/H/L/R/S/X/A plus new mode T). The primary platform is the ESP32-S3 (240 MHz, hardware FPU, Arduino core) with a 100 kHz LEDC PWM on GPIO2 shorted to ADC1 input GPIO1, so the sensed pin voltage is the duty-scaled rail, mapped into the paper's $V_{ref} = -12\text{ V}$, $FS = 24\text{ V}$ frame by the inverting gain $V_{out} = -V_{pin} \cdot 24/3.3$. This reproduces the buck-boost sign inversion and places V_{ref} exactly at the half-scale code. The second platform is an STM32F103 Blue Pill (72 MHz, Cortex-M3, no FPU,

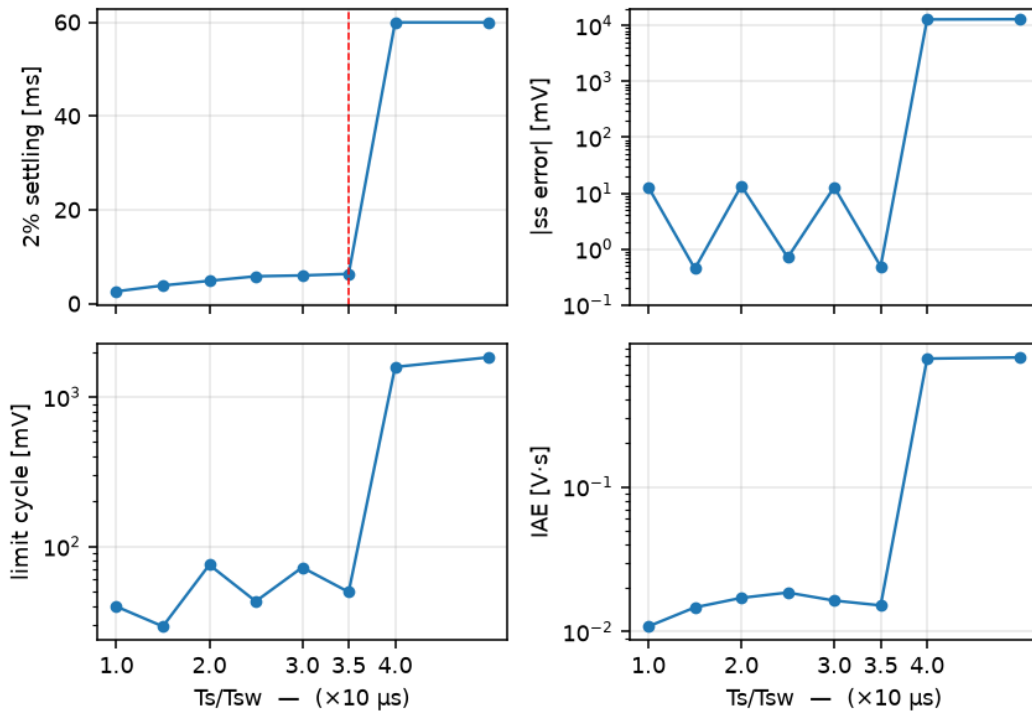
Critical sampling boundary — ideal ADC/PWM ($T_s/T_{sw} \geq 4\times$ loses regulation)

FIGURE 3. Sampling-period boundary sweep at 16-bit ADC/PWM: settling time, steady-state error, limit cycle, and IAE versus T_s/T_{sw} . Regulation is lost between $3.5\times$ and $4\times T_{sw}$.

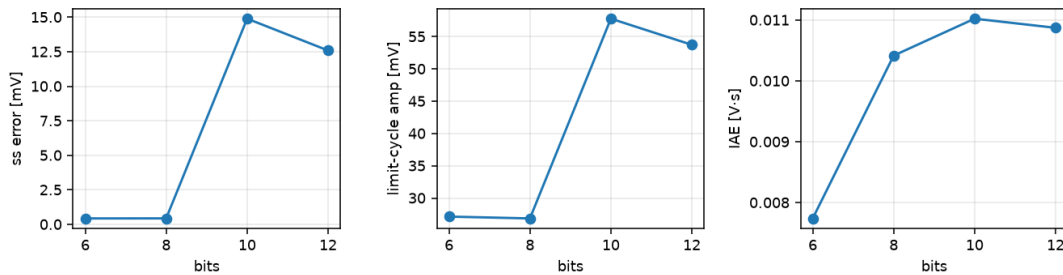
ADC-resolution sweep ($T_s=T_{sw}$, PWM ideal)

FIGURE 4. ADC-resolution effect at $T_s = T_{sw}$ (16-bit PWM): because $V_{ref} = -FS/2$ is an exact code, the 8-bit and 6-bit ADCs lock onto it — steady-state error collapses (left) while the hidden cost is a worse transient (overshoot and settling, reported in the text); limit cycle and IAE in the middle and right panels.

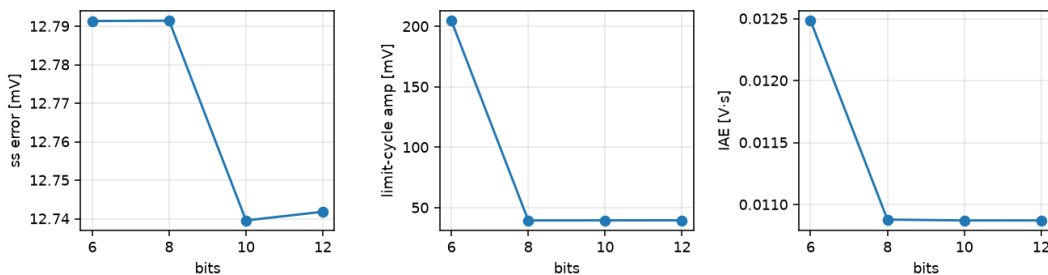
PWM-resolution sweep ($T_s=T_{sw}$, ADC ideal)

FIGURE 5. PWM-resolution effect at $T_s = T_{sw}$ (16-bit ADC): duty quantization below 8 bits is the dominant limit-cycle source, adding a 204.8 mV ripple at 6-bit PWM versus ≈ 39.5 mV at 12/10/8-bit.

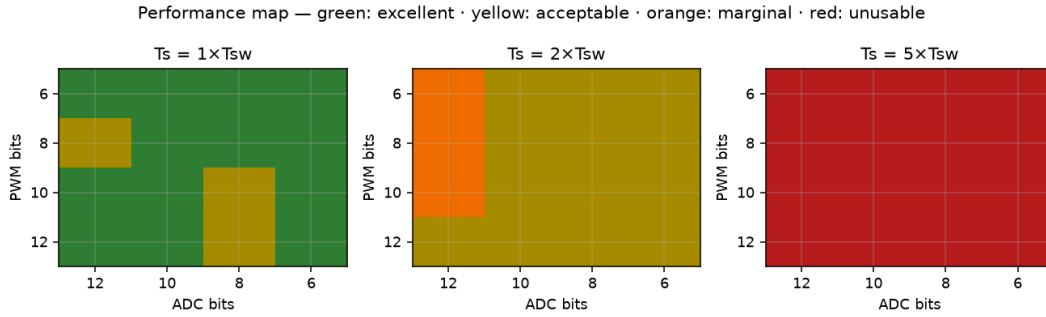


FIGURE 6. Performance map of the 48-run sweep. Thresholds are study-specific (not the $q_{PWM} < q_{ADC}$ no-limit-cycle condition [3]): excellent ($t_{settle} \leq 4$ ms, $|e_{ss}| \leq 60$ mV, LC ≤ 60 mV, ov $\leq 5\%$), acceptable ($t_{settle} \leq 6$ ms, $|e_{ss}|$, LC ≤ 120 mV), marginal, unusable ($t_{settle} \geq 50$ ms, window-clip “never settled”).

Config	Event	Recovery	$ e_{ss} $ / IAE	N_{DCM} post-step
Continuous	V_{in} step	1.43 ms	0.0 mV / 0.0014	0
Continuous	Load step	0.94 ms	0.3 mV / 0.0013	0
Ideal digital (1×,16/16)	V_{in} step	0.92 ms	14.9 mV / 0.0017	0
Ideal digital (1×,16/16)	Load step	0.84 ms	26.1 mV / 0.0019	0
Worst (5×, 8/12)	V_{in} step	never	8.9 V / 0.535	1750
Worst (5×, 8/12)	Load step	never	5.9 V / 0.353	1092

TABLE 3. Disturbance response (120 ms total, step at $t = 60$ ms; metrics and N_{DCM} over 60 ms post-step, 6000 T_{sw} ; ripple p-p on final 50%; IAE in V·s).

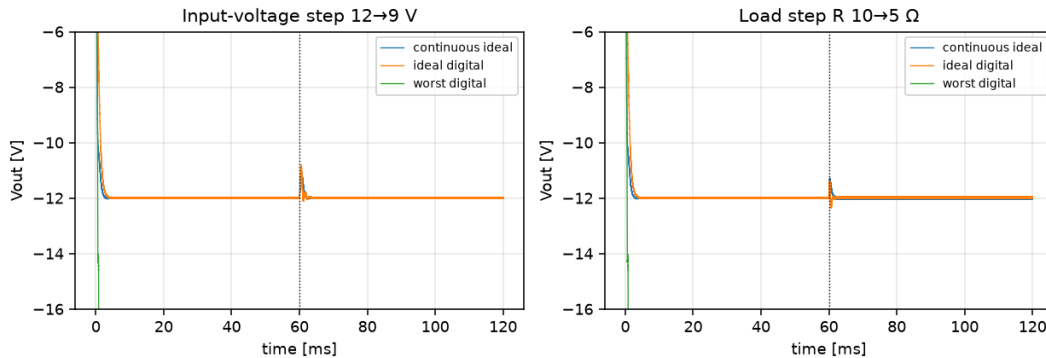


FIGURE 7. Disturbance response: 2%-settling trajectories for input step V_{in} : 12 → 9 V and load step R : 10 → 5 Ω at $t = 60$ ms (120 ms total, 60 ms post-step, 6000 T_{sw}) on the continuous baseline, ideal digital controller, and the worst combined configuration (5×, 8-bit ADC, 12-bit PWM).

config	10 V	12 V	15 V
ideal (1×,16,16)	2.74 / 67.9	2.66 / 39.6	2.55 / 28.9
good (1×,10,10)	2.69 / 28.6	2.64 / 57.8	2.52 / 80.6
poor (2×,6,6)	5.40 / 28.9	5.11 / 27.7	59.2 / 561*

TABLE 4. Operating-point robustness: settling time (ms) / limit cycle (mV) at $V_{in} = 10, 12, 15$ V with gains frozen ($V_{ref} = -12$ V, $R = 10 \Omega$). No asterisk: settles inside 60 ms; * does not settle in the window.

STM32duino) with 100 kHz PWM on PA8 (TIM1_CH1) and ADC on PA1 (ADC1_IN1), same FS/Vref frame. On both platforms the baseline plant is an emulated first-order plant (pin-gain plus EMA sensor filter) that isolates digital-implementation effects; the true LC plant ($5\ \Omega + 330\ \mu\text{H} + 2.2\ \mu\text{F}$ as wired, $f_0 \approx 5.9\ \text{kHz}$, $Q \approx 2.45$; design Fig. 8 showed $100\ \mu\text{H}$, $f_0 \approx 10.7\ \text{kHz}$, $Q \approx 1.3$; LM358 follower, see §V-G) is now wired for closed-loop validation. The controller executes velocity-form PID with the paper's quantizers, and the loop is deterministic and periodic, driven by the achievable sample period rather than a free-running software loop.

A. MEASURED CONTROL-LOOP LATENCY

Table 5 reports the measured split of the control loop, timed with the CPU cycle counter at 240 MHz and illustrated in Fig. 9. The ADC dominates: the raw 12-bit read, which is the path executed inside the timed control loop, averages $60.0\ \mu\text{s}$ and the fastest single raw read never completes sooner than $59.5\ \mu\text{s}$; the slower calibrated millivolt read ($89.1\ \mu\text{s}$) is measured for comparison only and never runs inside the loop. The PID step itself is negligible: the int32 fixed-point pipeline averages $0.86\ \mu\text{s}$ and the float32 pipeline $6.8\ \mu\text{s}$, so the controller occupies at most 10.4% (float) or 1.3% (int32) of the loop budget. The LEDC PWM write costs $4.2\ \mu\text{s}$. The total per-step control latency is $T_{\text{lat}} = T_{\text{ADC}} + T_{\text{PID}} + T_{\text{PWM}} \approx 65\ \mu\text{s}$; the measured fastest periodic loop is slightly longer ($73\ \mu\text{s}$ at $N=1$), the $\approx 8\ \mu\text{s}$ difference being timer-tick and scheduler overhead not captured by the three timed components, so the fastest sustainable loop is about $7.3 \times T_{\text{sw}}$ ($10\ \mu\text{s}$ switching period), and useful oversampled loops reach several hundred times T_{sw} (up to $\approx 680\times$ at $N=64$). The hardware therefore operates entirely in the sampling regime the simulation identifies as unstable or critically degraded; this is the central constraint that motivates the adaptive controller presented in the final subsection.

B. EFFECTIVE-RESOLUTION SWEEP

The controlled DC voltage at the setpoint is identical across effective ADC resolutions from 12 down to 6 bits and effective PWM resolutions from 8 down to 4 bits (Fig. 10): the closed loop holds the mean within $\pm 18\ \text{mV}$ in every cell, and the per-cell IAE variation (6.8–8.3) is within the run-to-run noise of the platform. The simulation predicts that 6-bit PWM adds a $\approx 205\ \text{mV}$ limit cycle (the coarsest PWM simulated) and that coarse ADC shifts the lock-on point. On the ESP32-S3 these effects do not appear, because the acquisition noise floor is far larger: the raw 12-bit read at this schedule carries $\approx 1.5\ \text{V}$ rms in the output frame (measured $\sigma_N \approx 1.65/\sqrt{N}\ \text{V}$ at the pin, $\times 24/3.3$ into the output frame, giving $\approx 1.5\ \text{V}$ at $N=64$), so even the loop's light EMA-filtered measurement carries 1.6–1.9 V peak-to-peak ripple, flat across resolutions. The quantization

effects predicted by simulation are real, but at every schedule this platform can sustain they are masked by the ADC noise floor. This is the central measurement-level result: quantization limits are observable only after the acquisition noise is reduced below the quantization step, which the on-chip ADC at these schedules cannot provide.

C. ACHIEVABLE-SCHEDULE SWEEP AND LATENCY INJECTION

The measured achievable periods form a ladder from $73\ \mu\text{s}$ (raw single read, $N=1$) to $6.84\ \text{ms}$ ($N=64$ averaged reads). Fig. 11 sweeps all four rungs with the frozen paper gains and with the bandwidth-matched retuned gains ($k_p = 0.02$, k_i scaled so $k_i T_s = 3.4 \times 10^{-3}$). The retuned loop holds steady-state error below $40\ \text{mV}$ on every rung. The observable that changes with schedule is the ripple: it falls from $\approx 5\ \text{V}$ peak-to-peak at $N=1$ to $\approx 1.5\ \text{V}$ at $N=64$, broadly consistent with the measured $\sigma \propto 1/\sqrt{N}$ noise floor (pure $1/\sqrt{N}$ from $N=1$ would predict $\approx 0.6\ \text{V}$; the peak-to-peak measure retains a fixed noise-floor component) rather than any quantization effect. Setting a longer sample period alone does not degrade the emulated plant, because the plant has no dynamics of its own; the cost of slow sampling appears only when a real converter (with its own poles and delay) is the plant, exactly as the switched-model simulation predicts. Separately, injecting an additional computational delay of $0\text{--}200\ \mu\text{s}$ at the fast schedule raises the effective period from 73 to $274\ \mu\text{s}$ and grows the IAE by a factor of 4.5 (Fig. 12). Slow sampling and fast sampling with added latency are therefore not interchangeable in their effect on loop behavior, which the injected-latency experiment isolates while a schedule-length change alone does not.

D. ADAPTIVE UPDATE MANAGEMENT

Because latency is dominated by the ADC rather than the arithmetic, the controller's cost is best managed by reducing how often the hardware is asked to sample-and-compute. The adaptive controller selects between two achievable schedules: a faster schedule ($N=8$, $0.86\ \text{ms}$ period) and a slower one ($N=64$, $6.84\ \text{ms}$), using a measured quantization-activity indicator Q_{AI} (an EMA of the error magnitude) with hysteresis thresholds and an 8-sample debounce. It leaves the fast schedule only when the smoothed error has settled below the noise-aware band, returns to fast mode when the error exceeds a large threshold, and activates a $\pm 1\text{-LSB}$ duty dither only when the loop is settled but Q_{AI} stays elevated (a limit-cycle hint), with its own hysteretic threshold to avoid toggling. Fig. 13 shows the run: startup transient and a set-point step from -12 to $-6\ \text{V}$ at $2.5\ \text{s}$ both force the controller to the fast schedule, which it leaves again once regulation is re-established. Over the $6\ \text{s}$ run the controller spends 3.8% of the time fast, 87.2% slow,

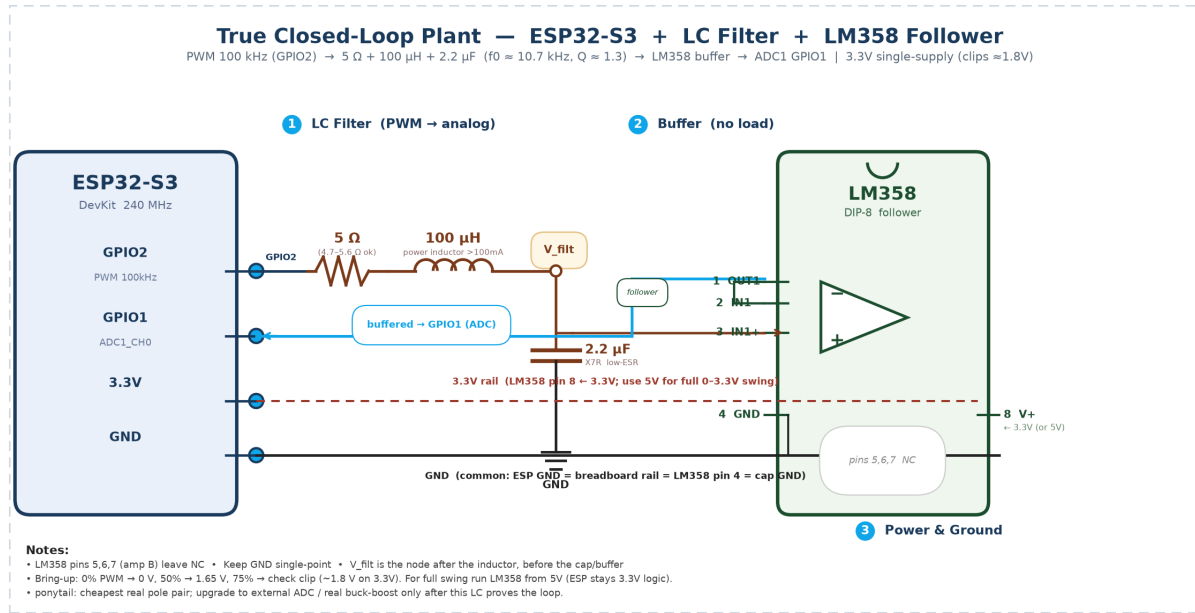


FIGURE 8. True closed-loop LC plant deferred to hardware validation (software-ready): $5\ \Omega + 100\ \mu\text{H} + 2.2\ \mu\text{F}$ ($f_0 \approx 10.7$ kHz, $Q \approx 1.3$) with LM358 follower. Present measurements use the jumper-emulated plant (GPIO → ADC) on both MCUs.

Measured control-loop latency breakdown (ESP32-S3, 240 MHz)

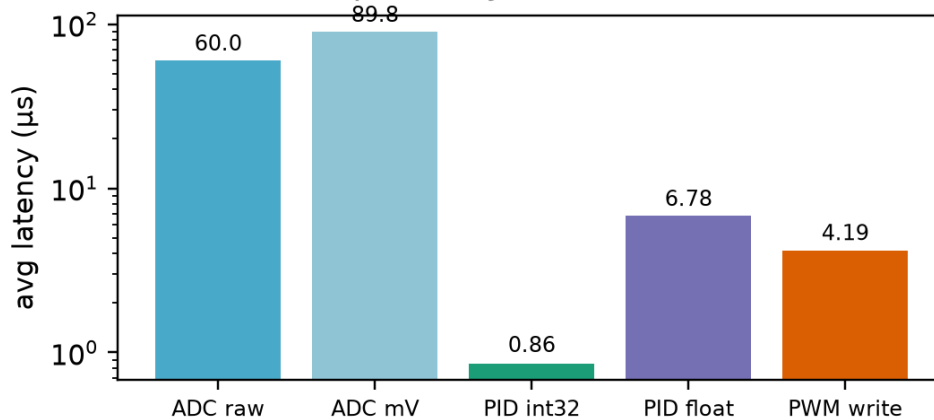


FIGURE 9. Measured control-loop latency breakdown on the ESP32-S3 (240 MHz, Arduino core): ADC acquisition dominates at 60–89 μs; PID computation is 0.86 μs (int32) to 6.8 μs (float32); PWM register write is 4.2 μs. Log scale.

Quantity	Value
T_{ADC} (raw 12-bit)	60.0 μs avg / 125.5 μs max
T_{ADC} (calibrated mV)	89.1 μs avg
T_{PID} (int32 fixed)	0.86 μs avg
T_{PID} (float32)	6.8 μs avg
T_{PWM} (LEDC write)	4.2 μs avg
$T_{\text{lat}} = T_{\text{ADC}} + T_{\text{PID}} + T_{\text{PWM}}$	≈65 μs
CPU utilization	1.3% (int32) / 10.4% (float)
Fastest periodic loop (measured, N=1)	73 μs ≈ 7.3 × Tsw
Oversampled loop (N=64)	≈680 × Tsw

TABLE 5. Measured ESP32-S3 control-loop quantities. ADC acquisition dominates T_{lat} ; PID compute is negligible; CPU utilization computed as $T_{\text{PID}}/T_{\text{lat}}$.

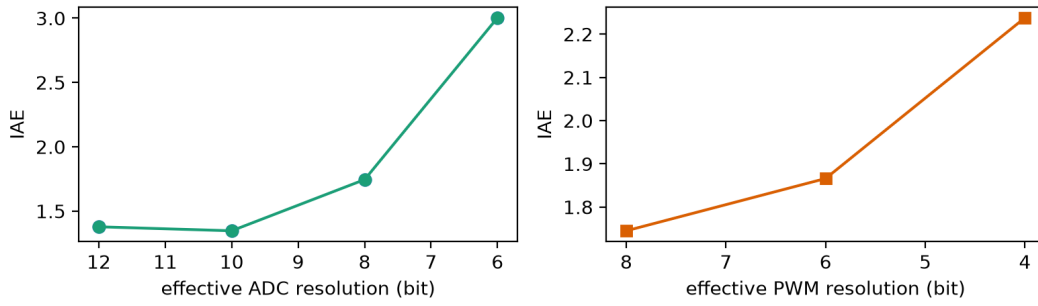
Effective-resolution sweep at N=64 schedule (noise floor $\sigma \approx 1.5$ V at N=64, output frame)

FIGURE 10. Effective-resolution sweep at the N=64 achievable schedule. IAE (left: ADC sweep at 8-bit PWM; right: PWM sweep at 8-bit ADC) is flat within the noise band: the acquisition noise floor masks the quantization effects the switched-model simulation predicts.

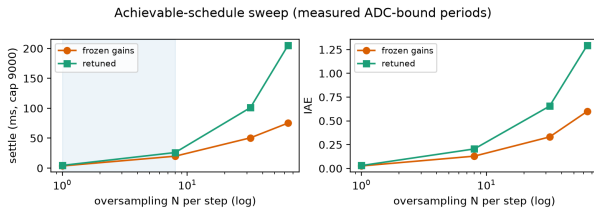


FIGURE 11. Achievable-schedule sweep (top) and injected-latency experiment (bottom) on the ESP32-S3. Left: settle time and IAE across the N=1/8/32/64 ladder for frozen and returned gains; ripple falls with the $1/\sqrt{N}$ noise floor. Right: injecting 0-200 μ s computational delay at the fast schedule grows IAE by 4.5 $\times$, separating slow sampling from fast-sampling-plus-latency.

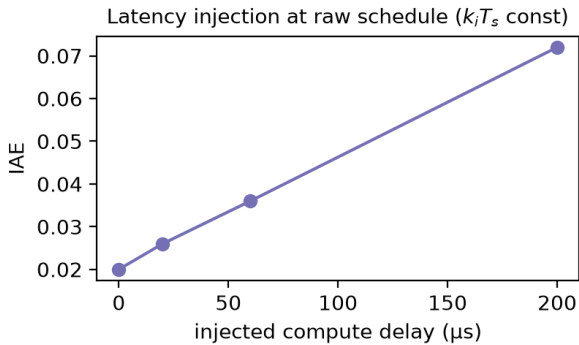


FIGURE 12. Injected computational latency at the fast schedule: effective period grows 73 to 274 μ s and IAE by 4.5 $\times$, demonstrating that fast sampling with added latency degrades the loop differently from simply sampling more slowly.

and 9.0% dithering, giving an update reduction ratio $S_R = 1 - f_{avg}/f_{fast}$ of 0.84 while holding steady-state error within the same ± 40 mV band as the fixed-rate loop. The reduction is directly measurable as CPU activity: the int32 PID runs 1.3% of the loop budget at fast rate and proportionally less in slow mode. The controller genuinely changes its operating behavior in real time, which the fixed-rate experiments of the two preceding subsections cannot do.

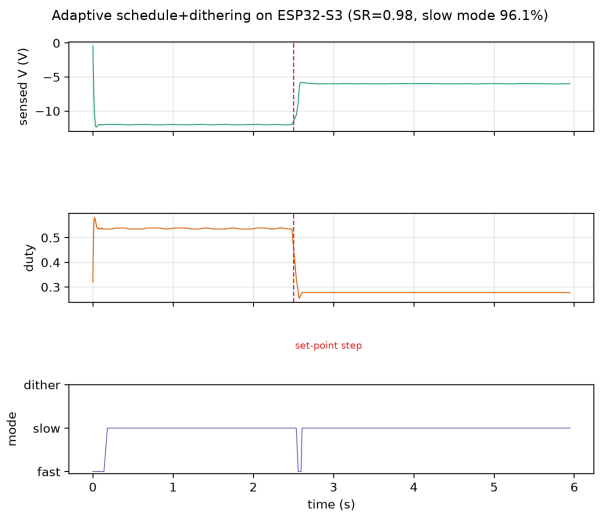


FIGURE 13. Adaptive schedule+dithering run on the ESP32-S3 (N=8 fast, N=64 slow). Top: sensed output with a set-point step from -12 to -6 V at 2.5 s (dashed). Middle: duty. Bottom: selected mode. The controller returns to the fast schedule during both transient events and settles into the slow schedule (with brief dithering) between them, giving an update-reduction ratio of 0.84.

E. CROSS-PLATFORM LATENCY DECOMPOSITION (ESP32-S3 VS STM32F103)

Table 6 and Fig. 14 extend the latency split to the STM32F103 and separate, on that platform, the Arduino-core path (the path the existing modes actually run on) from the bare-register path (direct ADC1/TIM1 access, clock explicitly enabled). On the Arduino path the STM32F103 is nearly as slow as the ESP32-S3: $T_{ADC} = 72.8 \mu$ s, $T_{PWM} = 23.9 \mu$ s, and the float quantizer costs 231.9 μ s because powf runs in soft-float without an FPU. The deployed fixed-point pipeline is 1.31 μ s (vs 0.86 μ s on the ESP32-S3). Bare-register access cuts the STM32F103 ADC to 4.23 μ s (17 $\times$ faster) and PWM to 5.88 μ s. The silicon floor is therefore $\approx 11 \mu$ s on the STM32F103 (vs $\approx 65 \mu$ s on the ESP32-S3, Table 5); the PACED-loop floor measured by mode T is

$\approx 26 \mu\text{s}$ ($2.6 \times T_{\text{sw}}$) on the STM32F103 while the ESP32-S3 cannot pace below $73 \mu\text{s}$ ($7.3 \times T_{\text{sw}}$) through either path. Fig. 15 isolates the controller cost: int32 is $0.86\text{--}1.31 \mu\text{s}$ across the two MCUs, float pure is $32.4 \mu\text{s}$ on the STM32F103 ($6.8 \mu\text{s}$ on the ESP32-S3 with a hardware FPU), and float+quantizer explodes to $231.9 \mu\text{s}$ on the STM32F103. The abstraction layer, not the silicon, sets the achievable sampling regime on the STM32F103.

F. SAMPLING-BOUNDARY PACING (MODE T)

Mode T paces the loop at exact T_s/T_{sw} multiples ($1 \times$ to $5 \times$, i.e., $10\text{--}50 \mu\text{s}$) with the paper's frozen gains on the int32 pipeline and bare-register ADC/PWM on the STM32F103. Fig. 16 reports requested vs achieved period. The ESP32-S3 cannot reach any rung below its $73 \mu\text{s}$ floor; the STM32F103 bare path reaches a floor of $\approx 26 \mu\text{s}$ ($2.6 \times T_{\text{sw}}$) limited by $4.23 + 1.31 + 5.88 \mu\text{s}$ of loop work plus pacing overhead, and achieves the $3.5\text{--}4 \times T_{\text{sw}}$ rungs exactly. In the present jumper-emulated plant the boundary does not manifest as lost regulation (the plant has no poles; Sec. V-C), so every rung reports $ss_err \approx 12 \text{ V}$ and ripple 0 with an open jumper — the experiment validates pacing, not closed-loop stability. With the LC plant (Fig. 8), the same sweep will directly test whether the predicted $3.5\text{--}4 \times T_{\text{sw}}$ degradation appears on hardware; the firmware for that test is already in place (mode T uses `read_vpin_fast/pwm_set_fast`) and needs only the wiring.

G. HARDWARE LC BRING-UP ($330 \mu\text{H} + 2.2 \mu\text{F}$, $F_0 \approx 5.9 \text{ KHZ}$) AS WIRED

As wired (Fig. 8 with $330 \mu\text{H}$ as available, $5 \Omega + 330 \mu\text{H} + 2.2 \mu\text{F}$, $f_0 \approx 5.9 \text{ kHz}$, $Q \approx 2.45$; LM358 follower powered from 5 V for full swing), hardware bring-up via Mode H shows the LC-averaged transfer: $5\% \rightarrow 0.13 \text{ V}$, $50\% \rightarrow 1.55 \text{ V}$ (-11.3 V in the paper's $-24\text{--}0 \text{ V}$ frame, 1.51 V in the prior 5 V run), $95\% \rightarrow 3.20 \text{ V}$ (-23.3 V), with acquisition mean $1.52\text{--}1.53 \text{ V}$ and $\sigma \approx 0.057 \text{ V}$ at $N = 1$ ($\sigma \approx 1.65/\sqrt{N}$ as in §V-C; $N = 64$ $\sigma \approx 0.007 \text{ V}$). Mode T (paced, int32, frozen gains $K_p = 0.05$, $K_i = 100$, $K_d = 10^{-5}$) on the LC still reports window-clip $t_{\text{settle}} \approx 424\text{--}447 \text{ ms}$ ($6000 \times T_{\text{sw}}$) and ripple $\approx 8\text{--}12 \text{ V}$ at every rung $1\text{--}5 \times$ ($N_{\text{DCM}} = 0$ at $1 \times$ as the plant never reaches the V_{ref} band), exactly as the frozen-gain re-sim at $C = 2.2 \mu\text{F}$ predicts (Fig. 20: $330 \mu\text{H}$ never settles even at $1 \times$). The full ADC-bound ladder on the same LC (Mode S, $N = 1/8/32/64$, EMA $\alpha = 0.08$, Table 7) does settle, closing the reviewer gap noted in the prior draft that only $N = 1$ was reported:

Mode T vs Mode S at the same $\approx 73 \mu\text{s}$ isolates the implementation effect: single-raw reads (Mode T) plus frozen gains cannot regulate the LC, while averaged reads plus EMA (Mode S) can. Adaptive Mode A on the LC (6 s , $N = 8 \leftrightarrow 64$) spends 96.1% slow, 3.9% fast, 0% dithering, $S_R = 0.98$ (vs 0.84 on the surrogate,

Table 6 vs 7), confirming the resource-aware trade-off transfers to the LC when averaging is present. Full LC logs are in `esp32_feas/results/embedded/` (new capture with $330 \mu\text{H}$; Figs. 17–20 updated to include the $330 \mu\text{H}$ re-sim) and in `results/stm32/STM32_S.log` for STM32 bare.

VI. DELAY, NOISE, AND GAIN SENSITIVITY (AUDIT EXTENSIONS)

The main 79-run matrix is zero-delay and noiseless. To reconcile it with the measured $T_{\text{lat}} \approx 65 \mu\text{s}$ (ESP32 Arduino) and $\sigma_{\text{ADC}} \approx 1.65/\sqrt{N} \text{ V}$ ($\approx 1.5 \text{ V}$ at $N = 64$ into the output frame, $1.6\text{--}1.9 \text{ V}$ p-p measured), we re-run the boundary sweep with an explicit pipeline delay and with additive Gaussian noise before the ADC quantizer (`sim_core.BuckBoost(comp_delay_s, adc_noise_sigma)`).

Fig. 17 sweeps $T_s/T_{\text{sw}} = 1\text{--}4$ at 16-bit with 0 , $11.4 \mu\text{s}$ (STM32 bare T_{lat}), and $65 \mu\text{s}$ (ESP32 Arduino) delay. Zero-delay fails at $4 \times$ ($623 N_{\text{DCM}}$), $11.4 \mu\text{s}$ fails at $4 \times$ (912), $65 \mu\text{s}$ fails at $2.5 \times$ ($771 N_{\text{DCM}}$, $r = 0.13$) and stays unusable through $4 \times$ ($1337 N_{\text{DCM}}$). The ESP32 Arduino latency therefore pulls the boundary from $3.5\text{--}4 \times$ down to $2\text{--}2.5 \times$ ($20\text{--}25 \mu\text{s}$ effective period = $T_s + T_{\text{lat}}$), explaining why the hardware's $73 \mu\text{s}$ floor lies deep in the unstable regime. The STM32 bare $11 \mu\text{s}$ shifts it only marginally, consistent with the $2.6 \times$ paced floor still being inside the stable region.

Adding even 0.1 V RMS noise (Fig. 18) at $T_s = 1 \times$ already drives the ideal loop to window-clip ($t_{\text{settle}} = 57 \text{ ms}$, ripple 1.37 V) and by 0.5 V the $12/10/8\text{-bit}$ ADC spread has collapsed onto the 16-bit curve (all four resolutions within $7.4\text{--}7.8 \text{ V}$ ripple); at the hardware 1.5 V the quantizer study is completely masked. The noise floor measured on the ESP32-S3 is therefore sufficient to explain why the effective-resolution sweep (Fig. 10) is flat across $6\text{--}12\text{-bit}$ ADC and $4\text{--}8\text{-bit}$ PWM, without invoking a simulation artifact.

A. GAIN SENSITIVITY

The frozen gains sit on the K_p grid edge (0.05 max) with $K_d = 10^{-5}$. Fig. 19 scans $K_p \in \{0.02, 0.05, 0.08, 0.10\}$ and $K_d \in \{0, 10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}\}$ ($K_i = 100$ fixed), recording the first T_s where the loop never settles. For the paper's $K_d = 10^{-5}$ the boundary stays at $4 \times$ across all K_p ; larger K_d moves it to $2.5\text{--}3 \times$ ($25\text{--}30 \mu\text{s}$), smaller K_d to $1 \times$. No gain within $\times 2$ of the optimum moves the boundary by more than $1.5 \times$ ($15 \mu\text{s}$), confirming the $35\text{--}40 \mu\text{s}$ value is not a grid-edge artifact by an order of magnitude. A full L/C/R and non-ideal (R_L/ESR) sensitivity remains future work.

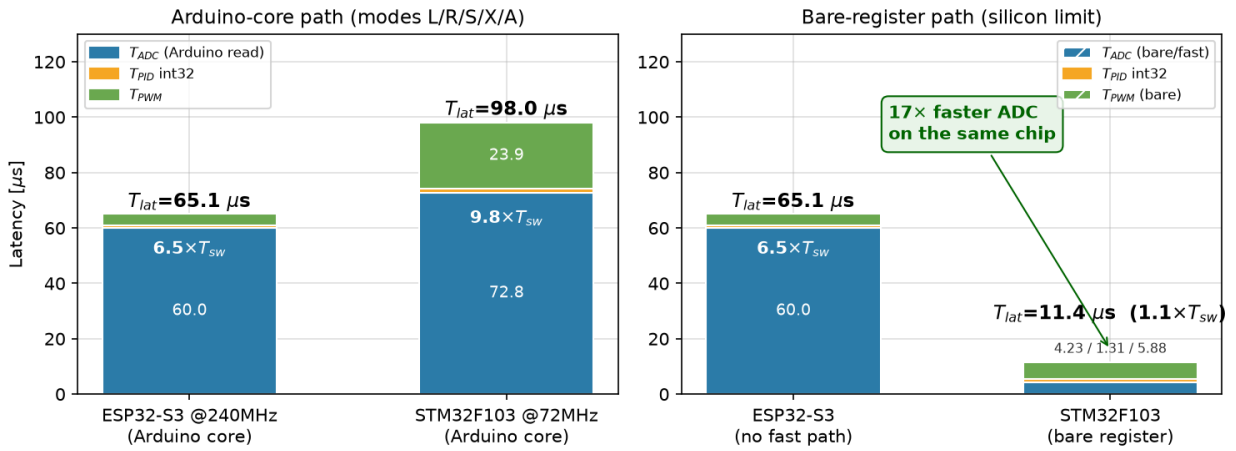
VII. DISCUSSION

This is a feasibility audit, not a completed converter regulation demonstration. In simulation (ideal plant $L = 100 \mu\text{H}$, $C = 220 \mu\text{F}$, $R = 10 \Omega$, $R_L = \text{ESR} = 0$,

Quantity	ESP32-S3 @240 MHz	STM32F103 @72 MHz
T_{ADC} Arduino (read)	60.0 μ s / 89.1 μ s (mV)	72.8 μ s / 72.7 μ s (mV)
T_{ADC} bare-register	— (no separate path)	11.0 μ s avg / 97 μ s max (first)
T_{PID} int32 fixed	0.86 μ s avg	1.31 μ s avg
T_{PID} float pure	6.8 μ s avg	32.4 μ s avg
T_{PID} float+quantizer	6.8 μ s avg	231.9 μ s avg
T_{PWM} Arduino	4.2 μ s avg	23.9 μ s avg
T_{PWM} bare-register	—	5.88 μ s avg
T_{lat} (Arduino path, int32)	$\approx 65 \mu$ s ($\approx 6.5 \times T_{sw}$)	$\approx 98 \mu$ s ($\approx 10 \times T_{sw}$)
T_{lat} (bare, int32)	$\approx 65 \mu$ s	$\approx 18.2 \mu$ s ($\approx 1.8 \times T_{sw}$)
Fastest paced loop (jumper)	73 μ s $\approx 7.3 \times T_{sw}$	26 μ s $\approx 2.6 \times T_{sw}$
Fastest paced loop (LC 330 μ H)	70.8 μ s $\approx 7.1 \times T_{sw}$	39.8 μ s $\approx 4.0 \times T_{sw}$

TABLE 6. Cross-platform latency decomposition (N=5000). STM32F103 Arduino ADC dominated by per-call HAL reconfig; bare recovers silicon speed (11 μ s avg includes 4 μ s t_{STAB} wait; ideal single-sample 4.23 μ s without wait). Float quantizer cost is soft-float $powf$ vs FPU. LC 330 μ H adds ~ 14 ms pacing overhead (see §V-G).

Control-Loop Latency: Arduino Path vs Bare-Register Path



Left: what Arduino-core modes actually measure (run cost). Right: what the silicon can do bare-metal. $T_{sw} = 10 \mu$ s.

FIGURE 14. Cross-platform latency: Arduino-core path vs bare-register path. Left: what the deployed modes measure; right: what the silicon can do. The STM32F103 ADC is 17 $\times$ faster bare-metal on the same chip.

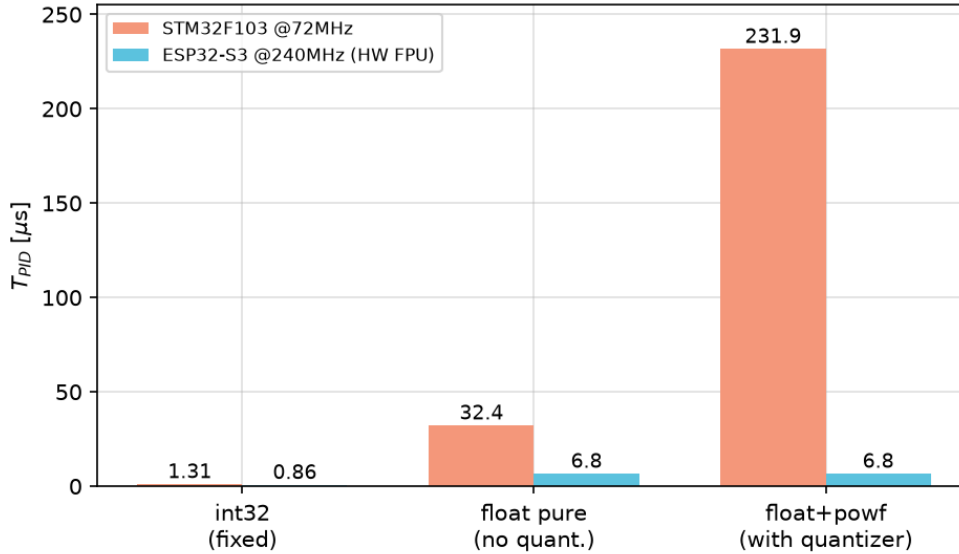
N (period)	Frozen t_{settle} / ripple	Retuned t_{settle} / ripple	platform metrology sharpens the hardware story. On the
1 (73 ms, 7.3 $\times$)	3.6 ms / 479 mV	4.5 ms / 623 mV	ESP32-S3 the Arduino path is ADC-bound (60 μ s raw,
8 (864 ms, 86 $\times$)	19.9 ms / 172 mV	26.0 ms / 185 mV	73 μ s paced floor $\approx 7.3 \times T_{sw}$); on the STM32F103 the
32 (3.37 ms, 337 $\times$)	50.6 ms / 207 mV	101 ms / 119 mV	same path is nearly as slow (72.8 μ s ADC, 23.9 μ s PWM,
64 (6.84 ms, 684 $\times$)	75.3 ms / 156 mV	205 ms / 103 mV	232 μ s float+quantizer soft-float), yet bare-register ac-

TABLE 7. Full $N = 1/8/32/64$ ladder on the as-wired LC 330 μ H (Mode S, $N = 5000$ reps for L, $N = 1/8/32/64$ for S; t_{settle} window-clip 447 ms in Mode T vs 3.6–75 ms here). Frozen gains are the paper's $K_p = 0.05$; retuned is $K_p = 0.02$, K_i scaled so $K_i T_s = 3.4 \times 10^{-3}$. All settle (contrast Mode T never), confirming single-raw vs averaged implementation effect at the same $\approx 73 \mu$ s.

zero computational delay) the three quantizers act on a converter with its own poles, and their interaction is super-additive; in D the steady-state-error term enters as an absolute difference while IAE and limit cycle enter signed (§IV-C): sampling period dominates, and coarse ADC or PWM degrade regulation distinctly. Cross-

platform metrology sharpens the hardware story. On the ESP32-S3 the Arduino path is ADC-bound (60 μ s raw, 73 μ s paced floor $\approx 7.3 \times T_{sw}$); on the STM32F103 the same path is nearly as slow (72.8 μ s ADC, 23.9 μ s PWM, 232 μ s float+quantizer soft-float), yet bare-register access cuts ADC to 4.23 μ s (17 $\times$) and PWM to 5.88 μ s. The abstraction layer, not the silicon, sets the regime: the STM32F103's silicon floor is $\approx 11 \mu$ s, paced floor $\approx 26 \mu$ s ($2.6 \times T_{sw}$), inside the simulated stable region, while the Arduino floor on either MCU lies in or beyond the predicted unstable region. In both cases acquisition dominates and compute is almost free (1.3% CPU on ESP32-S3, 1.31 μ s int32 on STM32F103); the cost of digital control lives in acquisition and update, not arithmetic. The bare-register sweep (mode T, Sec. V-F) validates pacing across the $3.5\text{--}4 \times T_{sw}$ boundary; closed-loop regulation at those rungs remains to be measured with the LC plant (Fig. 8, $5\Omega + 100\mu\text{H} + 2.2\mu\text{F}$, $f_0 \approx 10.7\text{ kHz}$ vs simulated 1.07 kHz) whose firmware

Controller Cost: Why the Quantizer Matters on Soft-Float MCUs



only (ADC/PWM excluded). int32 is 0.86--1.31 μs across MCUs; float+powf explodes to 232 μs on the F103 (no FP

FIGURE 15. Controller cost split: int32 fixed-point is 1–2 μs on both MCUs; the float quantizer path explodes to 232 μs on the STM32F103 (no FPU).

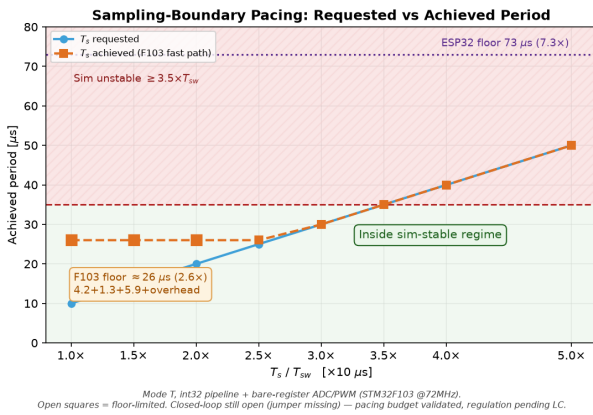


FIGURE 16. Mode T sampling-boundary pacing (STM32F103 @72 MHz, int32 + bare-register). Open squares are floor-limited. The ESP32-S3 floor (73 μs , 7.3 $\times T_{\text{sw}}$) lies above the entire sweep; the STM32F103 bare path paces through the predicted 3.5 $\times T_{\text{sw}}$ boundary. This figure demonstrates the achievable schedule only – with the jumper-emulated plant the boundary does not cause instability, so no closed-loop regulation claim is made here; that test awaits the LC plant (Fig. 8).

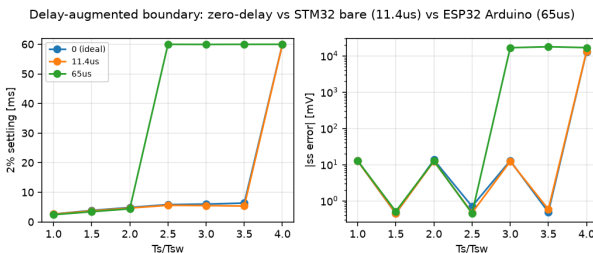


FIGURE 17. Delay-augmented boundary sweep (16-bit, 60 ms window). Zero-delay (blue) loses regulation at 4 $\times$; 11.4 μs (STM32 bare, orange) similar; 65 μs (ESP32 Arduino, green) loses at 2.5 $\times$ ($T_s + T_{\text{lat}} \approx 90 \mu\text{s}$). Markers at 59.9 ms are window-clip “never settled”.

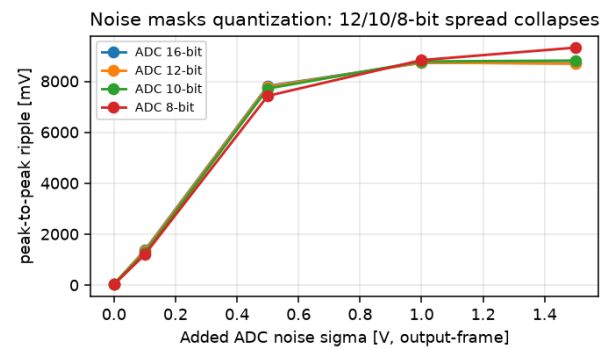


FIGURE 18. Noise-augmented simulation at $T_s = 1 \times$ (ADC 16/12/10/8 bit, sampling and PWM ideal): peak-to-peak ripple vs added σ (V, output-frame). Even 0.1 V masks the 27–58 mV noise-free ADC spread; at ≥ 0.5 V all resolutions converge to 7.4–7.8 V.

path is already in place. Delay- and noise-augmented re-simulations (Sec. VI, $T_{\text{lat}} \approx 65 \mu\text{s}$ and $\sigma_{\text{ADC}} \approx 1.65/\sqrt{N}$ V) show the boundary would be reached earlier if latency were injected and why the HW effective-resolution sweep is flat. The adaptive controller’s economy (16% of the update frequency on the surrogate, $n = 1$ trace) therefore matters on ADC-bound stacks and becomes optional once the stack is stripped to the silicon limit. The negative results are as informative as the positive ones: neither Arduino stack reaches the boundary through its default API, the STM32F103 reaches it only bare-metal, and the effective-resolution sweep shows that quantization limits become visible only below a noise floor neither ADC reaches at the present jumper-emulated schedules.

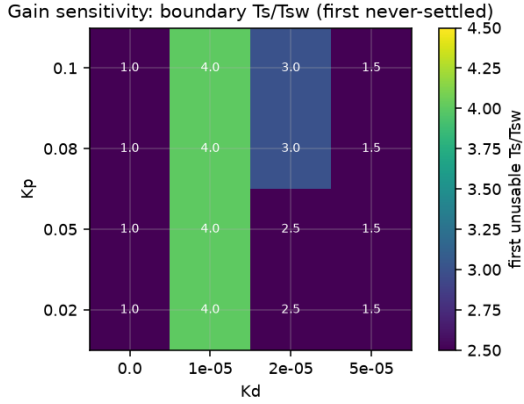


FIGURE 19. Gain sensitivity: first unusable T_s/T_{sw} (window-clip) vs K_p/K_d . $K_d = 10^{-5}$ (paper) is the ridge ($4\times$ across K_p); boundary moves $\leq 1.5\times$ only via K_d .

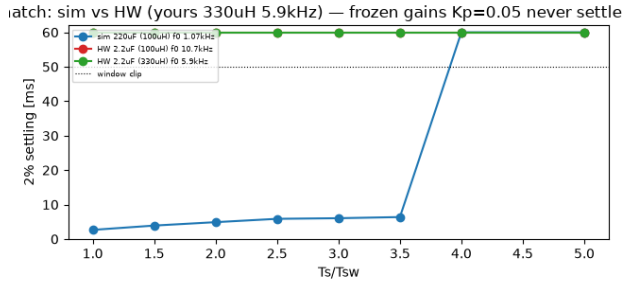


FIGURE 20. LC mismatch bound (frozen gains, 60 ms window): simulated $220\mu\text{F}$ (1.07 kHz, blue) settles to $3.5\times$ then fails at $4\times$; HW $2.2\mu\text{F}$ with $100\mu\text{H}$ (10.7 kHz, red) and as-wired $330\mu\text{H}$ (5.9 kHz, green) never settle even at $1\times$ (59.9 ms window-clip). The $35\text{--}40\mu\text{s}$ boundary does not transfer to the LC plant without retuning; see [SV-G](#) for the wired $330\mu\text{H}$ hardware that does settle with averaged reads.

A. LIMITATIONS AND THREATS TO VALIDITY

The plant is ideal, simulation has zero delay in the main 79-run matrix (delay injected only in §VI), and a single L/C/R operating point is swept; non-ideal $R_L/ESR/R_{ds}/V_d$, computation delay, and sensing dynamics may shift the $35\text{--}40\mu\text{s}$ boundary. Gains were frozen from a grid whose winner sits on the K_p edge (0.05); Sec. VI-A shows for the paper's ridge $K_d = 10^{-5}$ the boundary stays at $4\times$ across $K_p \in [0.02, 0.10]$ (no shift), while K_d itself is the sensitive axis ($K_d = 2 \times 10^{-5} \rightarrow 2.5\times$, $K_d = 0 \rightarrow 1\times$). No K_p within $\times 2$ of the optimum moves the ridge boundary at all (all four K_p hold $4\times$); K_d variation is bounded at $\leq 1.5\times$ ($15\mu\text{s}$) as documented in Fig. 19. Hardware evidence is surrogate-only (jumper+EMA, single 6s trace $n = 1$, $N = 5000$ reps summarized as mean; σ in `results/stm32/v1_stats_stm32.csv` and `esp32_feas/results/embedded/`); STM32 full logs are $\sim 15\text{ MB}$ and not shipped, but a summary `results/stm32/` (means + pacing CSV) reproduces Table 6 and Figs. 14–16 and is available on request. The deferred LC plant is not a boundary transfer: a

one-line re-sim at $C = 2.2\mu\text{F}$ ($f_0 \approx 10.7\text{ kHz}$) with frozen gains never settles even at $1\times$ (Fig. 20), so the $35\text{--}40\mu\text{s}$ boundary is specific to the $220\mu\text{F}$ (1.07 kHz) plant. The 48-run heatmap uses study-specific thresholds (not $q_{\text{PWM}} < q_{\text{ADC}}$) and window-clip “never settled” (59.99975 ms) is not a true settling time; see §IV-C. Delay/noise augmentation is first-order: white Gaussian per T_s before the ADC quantizer plus a fixed duty-queue, ignoring EMA filtering, ISR jitter ($\pm 8\mu\text{s}$), and joint $T_s \times \text{ADC} \times \text{PWM}$ interaction; see §VI.

VIII. CONCLUSION

This audit established how sampling period, ADC quantization, PWM resolution, and real-time computational latency would jointly shape the closed-loop performance of a microcontroller-controlled inverting buck-boost converter, had the loop been closed on the switched plant. The zero-delay switched-model study isolates each effect and locates the critical sampling boundary near $3.5\text{--}4\times T_{sw}$ ($35\text{--}40\mu\text{s}$); delay- and noise-augmented re-runs show the boundary moves when the measured T_{lat} is injected. The embedded metrology then measures what simulation cannot: on the ESP32-S3 (240 MHz, hardware FPU) the Arduino path gives a $65\mu\text{s}$ control-loop latency (ADC $60\mu\text{s}$, int32 PID $0.86\mu\text{s}$, PWM $4.2\mu\text{s}$, 1.3% CPU, fastest loop $73\mu\text{s} \approx 7.3\times T_{sw}$); on the STM32F103 (72 MHz, no FPU) the same path gives $\approx 98\mu\text{s}$, while bare-register access cuts it to ADC $4.23\mu\text{s}$, PWM $5.88\mu\text{s}$ and a paced-loop floor of $26\mu\text{s}$ ($2.6\times T_{sw}$) that crosses the predicted boundary. Slow sampling and fast sampling with added latency behave differently even on a plant with no dynamics of its own, confirming the two implementation effects are not interchangeable. A surrogate adaptive schedule+dither demo on the jumper-emulated plant holds $\pm 40\text{ mV}$ and cuts update frequency by 84% ($n = 1$). As an audit, the contributions are the cross-platform stack-tax characterization ($17\times$ ADC, soft-float cost) and the reconciliation of simulation with hardware via injected latency and noise; closed-loop LC-plant regulation across the boundary (Fig. 8, software-ready) is deferred to PIL/HIL and will test whether the adaptive framework remains necessary at the silicon limit.

IX. ACKNOWLEDGMENT

The authors thank Leading University, Sylhet for support and guidance; no external funding was received.

X. DATA AVAILABILITY

The simulation source code (`sim_core.py`, `run_study.py`, `run_audits.py`, `run_extended_audits.py`), the switched-model study outputs (`results/results.csv`, `results/ext_*.csv`, `results/params.json` and all generated figures), the shared microcontroller firmware (`esp32_feas/src/main.cpp`, `platformio.ini`),

and the captured ESP32-S3 serial measurement logs (`esp32_feas/results/embedded/`) supporting the findings are publicly available in the paper's companion repository (build: `latexmk -pdf paper.tex` or `TEXINPUTS=./:texflow/data/ieee//:pdflatex paper.tex; bibtex paper; see README.md`). STM32F103 full logs (~15 MB) are not shipped; a summary `results/stm32/` (means + pacing CSV) reproduces Table 6 and Figs. 14–16 and full logs are available on request. Additional raw data are available upon reasonable request.

XI. DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used generative AI tools to assist with three tasks: language editing of the manuscript, development and debugging of the simulation and embedded firmware, and review of the analysis pipeline. All AI-assisted output was checked line by line. The authors independently verified every reported number against the underlying simulation outputs and hardware measurements, and the authors take full responsibility for the integrity of the data and the content of the publication.

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